

Communication and Internet Technology: About Interconnecting Networks



Transmission Control Protocol (TCP) is the standard that defines how to establish and maintain a network conversation via which application programs can exchange data.

TCP works with Internet Protocol (IP), which defines how computers send packets of data to each other. Together, TCP and IP are the basic rules that define the Internet.

The transmission control protocol (TCP) layer, or layer 3, corresponds to layer 4 of Open Systems Interconnection, developed by International Standards Organization (popularly known as OSI/ISO protocol). This is the transport protocol. The layer operates in connection-oriented mode and guarantees delivery of data. TCP provides the service of breaking messages into datagrams at the source end, retransmitting any datagram that has been lost or acknowledged negatively, sequencing the datagrams and so on.

Service in the layer is guaranteed by the acknowledgement of receipt of data from the receiving side. If acknowledgement is negative, data is retransmitted, and if it is not received within a time period, known as

time-out period, retransmission of the packet is done. This layer also performs multiplexing and demultiplexing functions, if required.

The TCP layer makes TCP packs using the data received from application or higher layers. A TCP pack has many header fields, as shown in Figure 1.

TCP headers

Like IP, a TCP header is minimum 20 bytes in length. It has the following fields:

- A 16-bit source port number, which refers to a virtual circuit between two end-stations of communicating parties. The port is also called a socket or session. In a port or socket, more than one process can communicate over a session at any time between end-stations